

Speed Management & Van Safety

MODULE 02

FLEET SAFETY

Standard: Fleet & Field Safety
Format: Compliance Guide

⚠️ CRITICAL BUFFER: THE 4-SECOND FOLLOWING RULE

Commercial vans carry tools, racks, and heavy payloads that drastically increase momentum. Maintain at least a 4-second following interval under dry conditions. Extend to 6+ seconds at highway speeds or under heavy cargo load.

PHYSICS OF MOMENTUM & PAYLOAD

- **Kinetic Energy:** Doubling speed quadruples the kinetic energy brakes must dissipate.
- **Extended Stopping Distance:** Laden vans require significantly more distance to come to a complete halt.
- **Center of Gravity:** Racks and parts raise the vehicle roll-center, increasing turnover risk on rapid turns.

INCLEMENT WEATHER ADJUSTMENTS

- **Wet Asphalt / Rain:** Reduce speed by at least 1/3 (e.g., 60 mph down to 40 mph).
- **Packed Snow:** Reduce posted speed by at least 1/2.
- **Ice & Freezing:** Reduce to a crawl; tire friction is nearly zero.
- **Sight Line Limits:** Never drive faster than your clear sight distance allows.

DEFENSIVE IN-CAB TACTICS

- **Early Deceleration:** Scan ahead for brake clusters and slow down gradually to prevent panic stops.
- **Space Cushions:** Keep buffer space to the sides and avoid lingering in blind spots of larger vehicles.
- **Schedule Realism:** Speeding saves negligible time across urban routes while exponentially raising liability.

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Assessment: Verification Quiz
Requirement: 100% Mastery

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OPERATING PROTOCOLS

WORK ZONES & GRADE CONTROL

- **Work Zone Entry:** Scrub speed before entering cone tapers; expect narrow lanes and crew personnel.
- **Advisory Ramps:** Take off-ramps below posted advisory speeds to keep load balance.
- **Steep Descents:** Downshift gears early to use engine braking and avoid brake fade.

CRUISE CONTROL DIRECTIVES

- **Permitted Use:** Dry, clear, open highway conditions only.
- **Strict Ban:** Never engage cruise control in rain, standing water, ice, or snow.
- **Wheel Slip Risk:** Cruise control during low traction can trigger sudden drive-wheel loss and jackknife.

KNOWLEDGE VERIFICATION (NO-FAIL ASSESSMENT)

1. What is the minimum recommended following distance for a work van in dry conditions?

- | | |
|--------------|----------------------|
| A) 1 second | B) 2 seconds |
| C) 4 seconds | D) Half a car length |

2. By how much should you reduce your driving speed on wet roads during rain?

- | | |
|------------------|--------------------|
| A) Do not reduce | B) By at least 1/3 |
| C) By 5 mph only | D) Speed up |

3. Why do loaded commercial work vans require greater stopping distances?

- | | |
|-------------------------|--------------------------------------|
| A) Smaller brake pedals | B) Heavier weight increases momentum |
| C) Manual transmission | D) Tires have no tread |

4. When approaching an active road work zone, a driver must:

- | | |
|-------------------------------------|---|
| A) Accelerate to pass | B) Reduce speed early & watch for workers |
| C) Maintain top speed to first cone | D) Turn on high beams |

5. When is it unsafe to use cruise control in a work van?

- | | |
|------------------------|--|
| A) On open dry highway | B) In rain, snow, ice, or slick pavement |
| C) During daytime | D) On straight roads |